

AVICHAL BAWEJA

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SUMMARY

Software Engineer with full-stack experience building web applications, AI-powered systems, and automation solutions. Proficient in React, Python, FastAPI, and end-to-end product development. Strong background in backend development, database design, and cloud deployments in fast-paced startup environments with production-grade systems.

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Next.js, Responsive Design

Backend: Python, FastAPI, Django, Node.js, Express.js, REST APIs, GraphQL, Microservices

AI/ML: TensorFlow, PyTorch, OpenCV, YOLOv5, OCR/OMR, RAG Systems, AI Agents, LangChain

Tools & DevOps: Docker, Git, Linux, GCP, AWS, PostgreSQL, MongoDB, Redis, CI/CD

EXPERIENCE

Founding Engineer | *Datavedam*

Oct 2025 – Present

- Designed and developed an AI agent-based School ERP platform from concept to production with multiple integrated modules.
- Built RAG pipeline that scrapes CBSE circulars and converts them into vector embeddings for intelligent querying and decision support.
- Built OCR/OMR pipelines for document and exam processing; designed scalable backend services using Python and FastAPI.
- Built and managed data pipelines enabling efficient querying, filtering, and real-time decision-making for stakeholders.
- Managed DevOps workflows using Docker ensuring smooth CI/CD pipelines, production reliability, and zero-downtime deployments.

Business Analyst | *Purshotam Profiles Pvt. Ltd.*

Jan 2025 – July 2025

- Identified new business opportunities in the solar projects sector; prepared proposals and performance reports for client engagement.
- Collaborated with cross-functional teams to optimize sales processes, improve efficiency, and strengthen B2B client relationships.

Computer Vision Intern | *TechEagle Innovations Pvt. Ltd.*

Apr 2024 – May 2024

- Developed a CV-based drone system integrating YOLOv5 with real-time feedback control for stability, autonomous navigation, and obstacle avoidance.
- Designed an autonomous landing system in ROS/Gazebo using Python, OpenCV, and ArUco markers for precise coordinate-based landings.

IT Intern | *Sankalptaru Foundation*

Jan 2023 – Mar 2023

- Built predictive models using Python, Pandas, and scikit-learn for forecasting farm conditions; performed demand analysis with Matplotlib for agroforestry planning.

PROJECTS

Datavedam – AI Powered School ERP Platform | *Python, FastAPI, React, Docker*

- Implemented AI agents (Welfare Agent, AI Copilot); built QR-code based admission workflow handling 1000+ student enrollments.
- Integrated multiple APIs and backend services to automate school operations end-to-end with real-time dashboards.

Manufacturing Dispatch Management Portal | *React, FastAPI, PostgreSQL*

- Designed centralized web portal used by 15+ manufacturing plants for real-time dispatch management, logistics tracking, and GRN reconciliation.

Event Planner | *React, FastAPI, REST API, LLM*

- Built full-stack platform with integrated LLM-powered AI assistant for automated event descriptions, RSVP handling, and content generation.

Handwritten Text to LaTeX Converter | *Python, OCR, Image Processing, ML*

- Built OCR + LLM pipeline to extract handwritten notes and generate accurate LaTeX documents with optimized prompts for math expressions.

EDUCATION

Christ University

B.Tech. Computer Science & Engineering, Honors in AI & ML

Bengaluru, Karnataka

2021 – 2025

Delhi Public School

Senior Secondary School (CBSE)

Agra, Uttar Pradesh

2019 – 2021